Notes on Ricardian & HOS Model

1 Equalization via feasible transfers.

Take any measurable $A \subset [0,1]$ with $0 < |A| < 1$, and consider the budget-preserving perturbation

$$x_\varepsilon(t) = \begin{cases} x(t) + \varepsilon/p(t), & t \in A, \\ x(t) - \varepsilon/p(t), & t \in A^c, \end{cases} \qquad \varepsilon \in \mathbb{R} \text{ small.}$$

Then $\int_0^1 p(t)x_\varepsilon(t)\,dt = \int_0^1 p(t)x(t)\,dt = I$. Optimality of x^* implies the directional derivative at $\varepsilon = 0$ must vanish:

$$0 = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \int_0^1 \ln x_\varepsilon(t)\,dt = \int_A \frac{1}{x^*(t)}\frac{1}{p(t)}\,dt - \int_{A^c} \frac{1}{x^*(t)}\frac{1}{p(t)}\,dt.$$

Since this holds for *all* A, it follows that $\frac{1}{x^*(t)p(t)}$ is constant a.e., i.e.,

$$\frac{1}{x^*(t)\,p(t)} = \lambda \quad \text{a.e. on } [0,1]. \tag{1}$$

From (1), $x^*(t) = 1/(\lambda p(t))$. Imposing the budget constraint yields

$$I = \int_0^1 p(t)x^*(t)\,dt = \int_0^1 \frac{1}{\lambda}\,dt = \frac{1}{\lambda} \cdot 1,$$

so $\lambda = 1/I$ and hence

$$\boxed{x^*(t) = \frac{I}{p(t)} \quad \text{for a.e. } t \in [0,1].} \tag{2}$$

Remarks. (i) The result coincides with the n-good discrete case $x_i = I/(np_i)$ when the measure of the index set is 1. (ii) The argument avoided an inadmissible naive multiplier calculation by using measure-preserving feasible transfers; however, if one formalizes an infinite-dimensional Lagrangian under appropriate regularity (Slater, etc.), the same FOC $1/x = \mu p$ is obtained. (iii) Integrability of $\ln x^*$ requires $\int_0^1 |\ln p(t)|\,dt < \infty$, a mild regularity condition on prices.

2 Properties of the Cost Function

Consider a neoclassical production function $F_j : \mathbb{R}^2_+ \to \mathbb{R}_+$ (monotone, concave, differentiable). For input prices $(r, w) \in \mathbb{R}^2_{++}$ and output $y > 0$, define

$$C_j(r, w, y) := \min_{K,L \geq 0} \{ rK + wL \mid F_j(K, L) \geq y \},$$

and let $(\tilde{K}_j, \tilde{L}_j)$ denote the unique interior minimizer (conditional factor demands) under the stated assumptions.

(1) Monotonicity in output y

If $y' \geq y$, the feasible set $\{(K, L) : F_j(K, L) \geq y'\} \subseteq \{(K, L) : F_j(K, L) \geq y\}$, hence the minimum cannot decrease:

$$C_j(r, w, y') \geq C_j(r, w, y).$$

(2) Monotonicity in factor prices r, w

Fix y. If $r' \geq r$, then for any (K, L), $r'K + wL \geq rK + wL$. Taking minima over the same feasible set gives

$$C_j(r', w, y) \geq C_j(r, w, y),$$

and similarly for w.

(3) Positive homogeneity of degree one in (r, w)

For $\lambda > 0$,

$$C_j(\lambda r, \lambda w, y) = \min_{K,L} \{\lambda(rK + wL) \mid F_j(K, L) \geq y\} = \lambda\, C_j(r, w, y).$$

(4) Concavity in (r, w)

For fixed (K, L), the map $(r, w) \mapsto rK + wL$ is linear. The pointwise infimum over a family of linear functions is concave. Therefore $C_j(\cdot, \cdot, y)$ is concave on $\mathbb{R}^2_{++}$.

(5) Own-price monotonicity of conditional factor demands

Suppose C_j is differentiable (the non-differentiable case can be handled by subgradients). By concavity, the partial derivatives $\partial C_j/\partial r$ and $\partial C_j/\partial w$ are non-increasing in r and w, respectively. By Shephard's lemma below, these partial derivatives *equal* the conditional factor demands. Hence $\tilde{K}_j$ is non-increasing in r and $\tilde{L}_j$ is non-increasing in w.

(6) Shephard's lemma

Assume differentiability and interiority of the minimizer. By the envelope theorem,

$$\boxed{\frac{\partial C_j}{\partial r}(r,w,y) = \tilde{K}_j(r,w,y), \qquad \frac{\partial C_j}{\partial w}(r,w,y) = \tilde{L}_j(r,w,y).}$$

Sketch. Let $(\tilde{K}, \tilde{L})$ attain the minimum. For a small change in r, the value change is $dC = \tilde{K}\,dr + o(\|dr\|)$ because the indirect effect via $(\tilde{K}, \tilde{L})$ vanishes by the first-order conditions of the inner minimization. An analogous statement holds for w.

(7) Separation under constant returns to scale (CRS)

If F_j is homogeneous of degree one, define the unit-cost function

$$c_j(r,w) := \min_{K,L} \{\, rK + wL \mid F_j(K,L) \geq 1 \,\}.$$

Given any $y > 0$, by CRS we can write the feasibility condition as $F_j(yK', yL') \geq y$ iff $F_j(K', L') \geq 1$. Then

$$\begin{aligned} C_j(r,w,y) &= \min_{K,L} \{\, rK + wL \mid F_j(K,L) \geq y \,\} \\ &= \min_{K',L'} \{\, r(yK') + w(yL') \mid F_j(K',L') \geq 1 \,\} \\ &= y \min_{K',L'} \{\, rK' + wL' \mid F_j(K',L') \geq 1 \,\} \\ &= \boxed{c_j(r,w)\, y}. \end{aligned}$$

3 PPF Geometry, MRT, and Contract Curve

Let the economy be endowed with total capital $K > 0$ and labor $L > 0$. Two sectors $j \in \{1,2\}$ use (K_j, L_j) with $K_1 + K_2 = K$ and $L_1 + L_2 = L$. Outputs satisfy $y_j \leq F_j(K_j, L_j)$; F_j is neoclassical.

(1) The production possibility set is bounded, convex, and closed in $\mathbb{R}^2$

Let $\mathcal{A} := \{(K_1, L_1) \in [0,K] \times [0,L]\}$; this set is compact. The mapping

$$T : \mathcal{A} \to \mathbb{R}^2, \qquad (K_1, L_1) \mapsto \big(F_1(K_1, L_1),\ F_2(K - K_1, L - L_1)\big)$$

is continuous (composition of continuous maps). The feasible output set $\mathcal{Y} := \{(y_1, y_2) : \exists (K_1, L_1) \in \mathcal{A} \text{ s.t. } y_j \leq T_j(K_1, L_1)\}$ is the hypograph of T and thus closed; moreover, $T(\mathcal{A})$ is compact and hence bounded, implying $\mathcal{Y}$ is bounded.
Convexity: Take feasible (y_1, y_2) and (y_1', y_2') with associated allocations (K_1, L_1) and (K_1', L_1').

For $\theta \in [0,1]$, define $\bar{K}_1 = \theta K_1 + (1-\theta)K_1'$, $\bar{L}_1 = \theta L_1 + (1-\theta)L_1'$. By linearity of resource constraints, $(\bar{K}_1, \bar{L}_1)$ is feasible. By concavity of F_j,

$$F_1(\bar{K}_1, \bar{L}_1) \geq \theta F_1(K_1, L_1) + (1-\theta)F_1(K_1', L_1') \geq \theta y_1 + (1-\theta)y_1',$$
$$F_2(K-\bar{K}_1, L-\bar{L}_1) \geq \theta F_2(K-K_1, L-L_1) + (1-\theta)F_2(K-K_1', L-L_1') \geq \theta y_2 + (1-\theta)y_2'.$$

Hence $\theta(y_1, y_2) + (1-\theta)(y_1', y_2')$ is feasible; $\mathcal{Y}$ is convex.

(2) Marginal Rate of Transformation (MRT)

Consider an interior efficient allocation (PPF). A marginal reallocation of capital from sector 2 to sector 1, holding labor fixed, satisfies $dK_1 = -dK_2$ and $dL_1 = dL_2 = 0$. Then

$$dy_1 = \frac{\partial F_1}{\partial K_1} dK_1, \qquad dy_2 = \frac{\partial F_2}{\partial K_2} dK_2 = -\frac{\partial F_2}{\partial K_2} dK_1,$$

whence

$$-\frac{dy_2}{dy_1} = \frac{\partial F_2/\partial K_2}{\partial F_1/\partial K_1}.$$

An analogous argument with a labor reallocation yields

$$-\frac{dy_2}{dy_1} = \frac{\partial F_2/\partial L_2}{\partial F_1/\partial L_1}.$$

Thus, on the PPF,

$$\boxed{-\frac{dy_2}{dy_1} = \frac{\partial F_2/\partial K_2}{\partial F_1/\partial K_1} = \frac{\partial F_2/\partial L_2}{\partial F_1/\partial L_1}}.$$

(3) Location of the production contract curve in the Edgeworth box

Efficiency in production requires equality of MRTS:

$$\mathrm{MRTS}^1_{KL}(K_1, L_1) := \frac{\partial F_1/\partial L_1}{\partial F_1/\partial K_1} = \frac{\partial F_2/\partial L_2}{\partial F_2/\partial K_2} =: \mathrm{MRTS}^2_{KL}(K_2, L_2).$$

Under standard neoclassical assumptions (strict quasi-concavity and monotonicity), each sector's isoquants are strictly convex, and MRTS is strictly monotone in the capital-labor ratio K_j/L_j. Therefore the set of (K_1, L_1) satisfying $\mathrm{MRTS}^1_{KL}(K_1, L_1) = \mathrm{MRTS}^2_{KL}(K - K_1, L - L_1)$ is a continuous curve (the contract curve).

Let the *diagonal* be $\{(K_1, L_1) : K_1/K = L_1/L\}$. If the contract curve crossed the diagonal more than once in the interior, then by the intermediate value property and monotonicity of MRTS in K_j/L_j, one would obtain two distinct interior allocations with the same common MRTS but opposite strict orderings of factor intensities (i.e., $K_1/L_1 \gtrless K_2/L_2$ switched), contradicting the maintained factor-intensity ranking at common (supporting) factor prices.[1]

[1] Intuitively: with common supporting factor prices (r, w), cost-minimizing input ratios are pinned down in each sector; the sector that is relatively capital-intensive at those prices remains so at all efficient allocations supported by that price-hyperplane.

Hence the contract curve is either (i) exactly the diagonal or (ii) lies entirely on one side of it except possibly at the endpoints.

4 Factor Prices Equal Value Marginal Products

Claim. In an interior production equilibrium,

$$\boxed{r = p_j \frac{\partial F_j}{\partial K_j}, \qquad w = p_j \frac{\partial F_j}{\partial L_j} \quad (j = 1, 2).}$$

Proof. Consider the unit-cost minimization for good j:

$$c_j(r, w) := \min_{K,L} \{\, rK + wL \mid F_j(K, L) \geq 1 \,\}.$$

The Lagrangian is $\mathcal{L} = rK + wL - \mu(F_j(K, L) - 1)$ with $\mu > 0$. First-order conditions (necessary and sufficient by convexity) are

$$r = \mu \frac{\partial F_j}{\partial K}, \qquad w = \mu \frac{\partial F_j}{\partial L}, \qquad F_j(K, L) \geq 1, \;\; \mu(F_j(K, L) - 1) = 0.$$

Under CRS, Euler's theorem gives $F_j = \frac{\partial F_j}{\partial K} K + \frac{\partial F_j}{\partial L} L$. Multiply the FOCs by K, L and sum:

$$rK + wL = \mu \left(\frac{\partial F_j}{\partial K} K + \frac{\partial F_j}{\partial L} L \right) = \mu F_j(K, L) = \mu.$$

At the optimum with $F_j = 1$, the minimized value is $c_j(r, w) = rK + wL = \mu$. In competitive long-run equilibrium, zero-profit implies $p_j = c_j(r, w)$. Substituting $\mu = p_j$ into the FOCs yields the result:

$$r = p_j \frac{\partial F_j}{\partial K}, \qquad w = p_j \frac{\partial F_j}{\partial L}.$$

□

5 Homothetic Preferences ⇒ Linear Engel and Separation

Claim. If preferences are homothetic, then for $j = 1, 2$,

$$\boxed{x_j = D_j(p_1, p_2, I) = \varphi_j(p_1, p_2)\, I.}$$

Proof 1 (Direct definition). Homotheticity means that for all $\alpha > 0$, $x(p, \alpha I) = \alpha x(p, I)$. Setting $\alpha = 1/I$ gives

$$x_j(p, I) = I \cdot x_j(p, 1) =: \varphi_j(p)\, I,$$

where $\varphi_j(p)$ depends only on prices. Hence demand separates into a purely price-dependent part and income linearly.

Proof 2 (Duality). For homothetic preferences, the expenditure function is linear in utility: $e(p,u) = a(p)\,u$ for some positive price index $a(\cdot)$. The indirect utility is $v(p,I) = I/a(p)$. By Roy's identity,

$$x_j(p,I) = -\frac{\partial v/\partial p_j}{\partial v/\partial I} = I \cdot \frac{\partial \ln a(p)}{\partial p_j} =: \varphi_j(p)\,I,$$

again yielding the separation.